TT-P-1181A
Amendment-1
January 24, 1972

FEDERAL SPECIFICATION

PAINT, STYRENE-ACRYLATE SOLVENT TYPE, TINTS AND DEEP TONES
(FOR EXTERIOR MASONRY)

This amendment, which forms a part of Federal Specification TT-P-1181A dated May 4, 1968 was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

PAGE 1

Under Federal Specifications: Delete the following:

TT-P-442 - Pigment, Titanium Dioxide (For Protective Coatings).

Add at the bottom of the page the following:

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 476 - Titanium Dioxide Pigments

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

PAGE 2

Paragraph 3.1 line 4. Add "The solvents or solvent system used in the process shall comply with Air Pollution Regulations "Rule 66" 1/. A certificate of compliance to this effect is necessary."

Paragraph 3.2.1, line 2, change "TT-P-442" to "ASTM D 476".

Bottom of page 4. Insert new footnote 1/ as follows:

1/ Information on "Rule 66" may be obtained from Air Pollution Control District, County of Los Angeles, California 90001.

PAGE 4

Paragraph 4.4.5 through 4.4.6. Delete in entirety and substitute:

FSC 8010

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:18Z
Check the source to verify that this is the current version before use.

4.4.5 Determination of styrene-acrylate resin

Apparatus: Supercentrifuge
Centrifuge
Double beam grating infrared recording spectrophotometer
(spectral resolution 3 cm or better over entire range at slow scan)
Rock salt plates for use in spectrophotometer
Vacuum oven
Drying oven
Steam bath
Soxhlet extraction apparatus- 500-ml erlenmeyer flask-24/40 neck
Hot plates

4.4.5.1 Determination of chlorinated paraffins and additives (n-pentane extractables).

Weigh, by difference to the nearest mg., 50 to 55g. of well mixed sample, using a weighing bottle and add to 100g. of sodium chloride C.P. in a 500-ml glazed porcelain casserole. Mix well and evaporate most of the solvent by heating the casserole on top of the steam bath (not directly over an opening) for 6 hrs. Sir mixture occasionally during evaporation. By using an infrared lamp in addition to the steam bath, solvent evaporation time can be reduced from 6 to 2 hrs. Remove the mixture from the casserole while still slightly damp damp and grind as finely as possible in a mortar. Dry in the casserole overnight in a regular drying oven at 60 - 70°C. There should be no odor of solvent. Regrind to a powder as fine as salt and transfer to a Soxhlet thimble (43 x 123 mm) and cover with a plug of glass wool. Extract with 400 ml of tech. grade n-pentane under vigorous reflux for eight hours, or if more convenient, let run overnight (see note). Use a Soxhlet extraction apparatus equipped with a calcium chloride tube and five Berl saddles as boiling stones in the 500-ml erlenmeyer flask.

At the conclusion of the extraction period, remove the extraction thimble and rinse with n-pentane. Rinse the Soxhlet with n-pentane. Transfer the warm pale yellow n-pentane extract and rinsings to a tared 400-ml beaker containing three Berl saddles. Rinse the extraction flask with n-pentane and add to the 400-ml beaker. The film remaining on the bottom and sides of the flask is low molecular weight styrene-acrylate resin and is not to be combined with the n-pentane extract which contains the chlorinated paraffins. Evaporate the n-pentane extract on a cooler portion of the steam bath. When the n-pentane has been evaporated, leave the beaker directly over the opening at full steam to remove any traces of solvent. Dry in a vacuum oven to constant weight at 60-70°C. This requires overnight (16 hrs.) drying. Weigh beakers and calculate weight of n-pentane extract.

$$\frac{\text{Net wt. of n-pentane extract} \times 100}{\text{wt. of sample}} = \text{Percent chlorinated paraffins and additives}$$

Source: https://assist.dla.mil -- Downloaded: 20261O-26T12:18Z
Check the source to verify that this is the current version before use.

Notes

1. The salt-paint mixture is evaporated on top of the steam bath to prevent any decomposition of the styrene-acrylate resin. It was found that white paints containing styrene resins became gray when placed in a casserole directly over the steam opeining, owing to decomposition. If, on the other hand, the same paint was evaporated on top of the steam bath, it remained white. Temperature measurements showed 73°C on top and 100°C below the steam bath opening.

2. The purpose of evaporation is to eliminate the original paint solvents. These must be eliminated in order to reduce as much as possible the solubility of the styrene-acrylate resin in the n-pentane.

3. The salt-paint mixture should be removed from the casserole while still damp. If allowed to dry, it acquires a rock-like hardness, thus making removal and grinding laborious and time-consuming.

4. It was found that 98 percent of the n-pentane extract obtained in a 22 hr. Soxhlet extraction is achieved in 8 hours.

4.4.5.2 Determination of total benzene extractables (non-volatile vehicle) (styrene-acrylate resin plus chlorinated paraffins 40 percent and 70 percent in 1:1 ratio)

Weigh 50 to 55g. of well-mixed sample to the nearest mg., using a weighing bottle and wash with benzene into a supercentrifuge bowl of 300 ml capacity. Mix well and dilute with benzene to approximately 80 percent of capacity of the bowl. Benzene may be easily lost during centrifugation. Make certain that the gasket is in good condition and the cover sealed tightly. Centrifuge for 1/2 hr. at 50,000 rpm.

Cool the bowl externally before opening. Decant the liquid into a tared 250-ml beaker containing three Berl saddles (as boiling stones) and evaporate to dryness on a steam bath. Extract the material remaining in the bowl by throughly mixing with benzene and proceeding as before. Decant into a second tared 250-ml beaker containing three Berl saddles and evaporate to dryness. Generally, the first extract is free of pigment and the second extract has a small amount. The presence of pigment in the decantate is indicated by a cloudy benzene solution.

When pigment is present in either extract, remove it by the following procedure. Evaporate the benzene extract to dryness on the steam bath, take up in benzene and transfer to two 100-ml conical-shaped centrifuge tubes. Centrifuge at 2000 rpm for 45 minutes and decant into the original beaker and again evaporate on the steam bath. If both extracts contain pigment, centrifuge the more concentrated extract first. Use the same centrifuge tubes for the second extract to eliminate possible mechanical loss.

Dry beakers on the steam bath until only a faint odor of solvent remains. (By leaving a slight amount of solvent in the extract, the extract assumes a honeycomb structure upon vacuum drying and is therefore

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:18Z
Check the source to verify that this is the current version before use.

more easily dried). Dry beakers in a vacuum oven at 60 - 70°C and 30 in. of mercury to constant weight (approximately 32 hrs.). Weigh beakers and calculate total weight of extractables.

$$\frac{\text{Total wt. of extracts} \times 100}{\text{wt. of sample}} = \text{percent total benzene extractables}$$

4.4.5.3 Calculation of resin content.

Percent styrene-acrylate resin in paint sample = percent total benzene extractables - percent n-pentane extractables.

4.4.6 Identification of resin from vehicle.

Apply a few drops of the benzene extract (see 4.4.5.2) to a sodium chloride disc and dry at 60°C for 30 minutes. When the film is of adequate thickness as seen by a trial infrared spectrum, dry disc in vacuum oven until the benzene is completely removed as shown by absence of an absorption band at 14.8 μ. Compare the infrared spectrum with that shown in Figure 1. There should be a close resemblance.

PAGE 6

Figure 1. Delete caption and substitute the following:

Infrared spectrum of styrene-acrylate resin, with chlorinated paraffin 40 percent and chlorinated paraffin 70 percent in the proportion 2:1:1 based on the formulation in Table III.

Figure 2. Delete entirely.

PAGE 7

TABLE III, lines 12 and 13, Delete:

"Solvesso 100	203	199
Mineral spirits	203	199"

and substitute:

"Rule 66 replacement solvent	406	398"

PAGE 8

TABLE V

Deep-tone blue
Lines 12 and 13, Delete:

"High flash naphtha	193
Mineral spirits	193"

and substitute:

"Rule 66 replacement solvent	386"

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:18Z
Check the source to verify that this is the current version before use.

TABLE V. (Continued)

Deep-tone brown
Lines 10 and 11, Delete:

"Hi flash naphtha	194
Mineral spirits	194"

and substitute:

"Rule 66 replacement solvent	386"

Sun-yellow
Lines 10 and 11, Delete:

"Hi flash naphtha	203
Mineral spirits	203"

and substitute:

"Rule 66 replacement solvent	406"

TABLE V (Continued)

Deep-tone green
Lines 12 and 13, Delete:

"Hi flash naphtha	190
Mineral spirits	190"

and substitute:

"Rule 66 replacement solvent	380"

Deep-tone red
Lines 12 and 13, Delete:

"Hi flash naphtha	200
Mineral spirits	200"

and substitute:

"Rule 66 replacement solvent	400"

TABLE VI

Lines 3 and 4. Delete:

"Mineral spirits	65
Solvesso 100 (R)	65"

and substitute:

"Rule 66 replacement solvent	130"

Lines 10 and 11, Delete:

"Mineral spirits	134
Solvesso 100 (R)	134"

and substitute:

"Rule 66 replacement solvent	268"

☆ U. S. GOVERNMENT PRINTING OFFICE : 1972 O - 482-193 (3381)

Source: https://assist.dla.mil -- Downloaded: 20261O-26T12:18Z
Check the source to verify that this is the current version before use.

1 a 4-1/2 inch rubber-covered roller. down. The tape shall be pressed down with two passes o